Rhenium

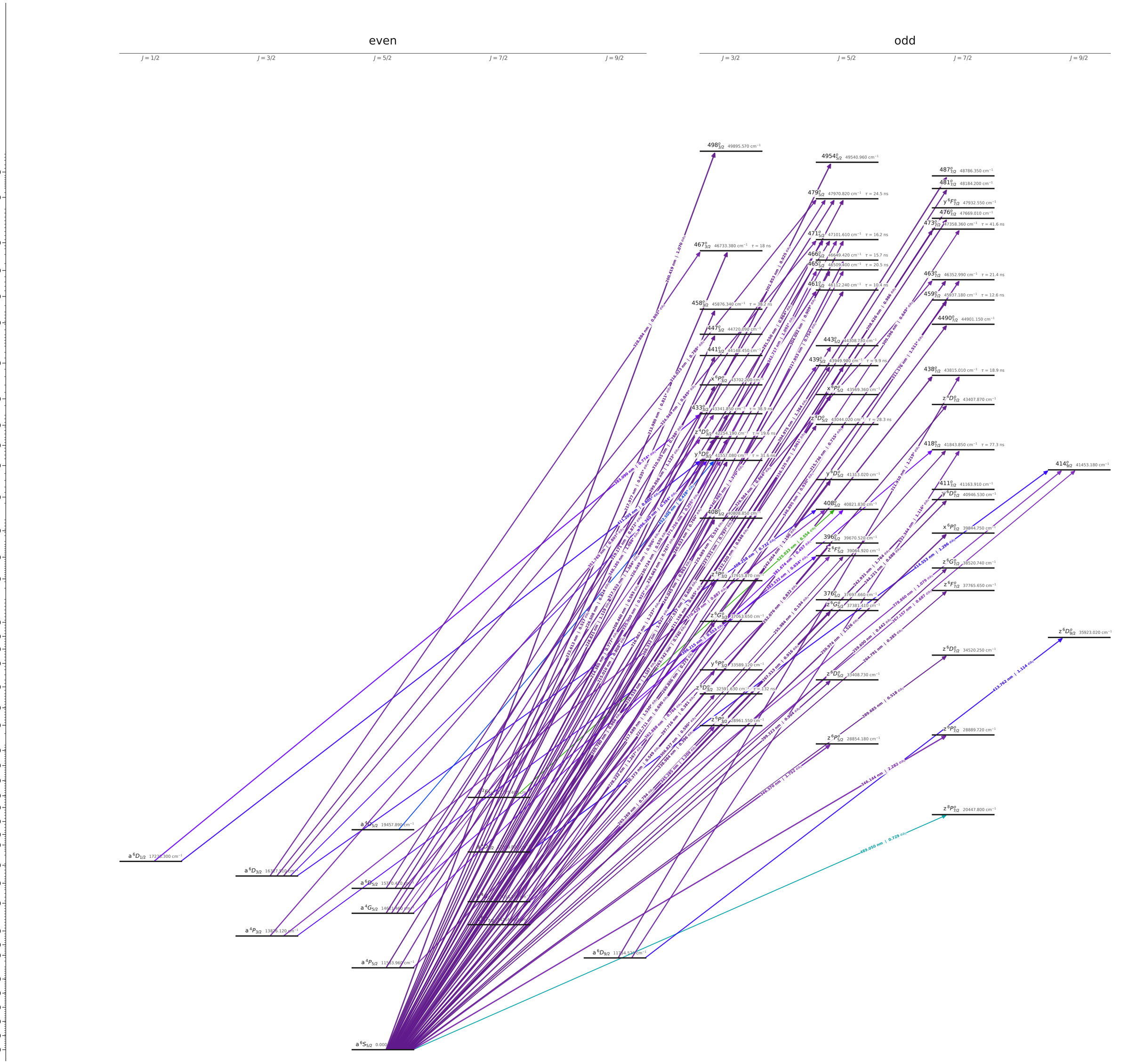
Re (Re I). Strongest classified transitions below 2 μm (NIST): 67 levels, 90 lines.

Re⁺ ionisation limit 63181.60 cm^{-1} = 7.83352 eV (158.274 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\|\langle J\|er\|J'\rangle\|$ (ea_0)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a $^6S_{5/2}$ – 498$^o_{3/2}$	200.4186	200.3538	1495.8316	1.070	7.210e+07	–	–	
a $^6S_{5/2}$ – 4954$^o_{5/2}$	201.8532	201.7881	1485.2006	0.925	3.510e+07	–	–	
a $^6S_{5/2}$ – 487$^o_{7/2}$	204.9754	204.9097	1462.5780	1.364	5.470e+07	–	–	
a $^6S_{5/2}$ – 467$^o_{3/2}$	213.9798	213.9124	1401.0315	0.811*	3.400e+07	8.842	61.1	
a $^6S_{5/2}$ – 461$^o_{5/2}$	216.8622	216.7941	1382.4102	1.237*	5.070e+07	15.3	52.6	
a $^6S_{5/2}$ – 459$^o_{7/2}$	217.6886	217.6204	1377.1620	1.530*	5.750e+07	12.63	72.4	
a $^6S_{5/2}$ – 438$^o_{7/2}$	228.2323	228.1619	1313.5410	1.267*	3.420e+07	8.421	64.8	
a $^6S_{5/2}$ – x $^6P^o_{3/2}$	228.8214	228.7509	1310.1590	1.119	5.290e+07	–	–	
a $^6S_{5/2}$ – x $^6P^o_{5/2}$	229.5191	229.4484	1306.1765	1.107	3.420e+07	–	–	
a $^6S_{5/2}$ – y $^6D^o_{5/2}$	242.0544	241.9809	1238.5332	1.190	3.370e+07	–	–	
a $^6S_{5/2}$ – 411$^o_{7/2}$	242.9312	242.8575	1234.0630	1.764	5.500e+07	–	–	
a $^6S_{5/2}$ – 408$^o_{3/2}$	245.0449	244.9707	1223.4185	0.963	3.190e+07	–	–	
a $^6S_{5/2}$ – x $^6P^o_{7/2}$	250.9741	250.8985	1194.5156	1.526	3.730e+07	–	–	
a $^6D_{7/2}$ – 466$^o_{5/2}$	308.3321	308.2426	972.3037	1.622*	3.030e+07	10.14	47.6	
a $^6D_{7/2}$ – 439$^o_{5/2}$	336.3253	336.2287	891.3765	2.061*	3.770e+07	16.08	37.3	
a $^6S_{5/2}$ – z $^6P^o_{7/2}$	346.1439	346.0448	866.0920	2.282	3.180e+07	–	–	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).